Projective blowups: a formal and multicentered proof

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Abstract

We give a machine-checked proof, in Lean 4, of the universal property of multicentered blowups: given a finite family $\boldsymbol{Z}$ of closed subschemes of a scheme $\boldsymbol{X}$, presented on each chart by ideals, there is a scheme $\mathbf{Bl}_{\boldsymbol{Z}}\,\boldsymbol{X}$ over $\boldsymbol{X}$, terminal among $\boldsymbol{X}$-schemes on which every member of the family becomes an effective Cartier divisor. To the best of our knowledge this is the first formalisation of blowups, single- or multicentered, in any theorem prover. The proof is organised around the universal property of the multicentered dilatation of a ring: the initial algebra in which each ideal of a finite family becomes generated by a prescribed non-zero-divisor. The comparison isomorphisms on overlaps, the cocycle identity, and the independence of the chosen presentation are obtained from the uniqueness part of this universal property. We give the exact theorem statement as formalized, the proof architecture, and the main new mathematical ingredients, together with the corresponding Lean code.

Keywords: blowups, dilatations, formalization of mathematics, Lean 4, multi-graded Proj, algebraic geometry, Cartier divisors

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1 Introduction

Blowing up a scheme along a closed subscheme is one of the foundational operations in algebraic geometry. The classical blowup solves a lifting problem: any X-scheme on which the centre becomes a Cartier divisor admits a unique morphism to the blowup.

For a finite family of centres, the multicentered blowup is the universal space on which all the centres become Cartier simultaneously. This paper establishes the universal property of multicentered blowups with a machine-checked proof in Lean 4 [3]. To the best of our knowledge this is the first formalisation of blowups, single- or multicentered, in any theorem prover.

The key tool is multi-centered dilatations and their universal property, which turns ideals into principal ideals. The global blowup is obtained by gluing dilatation charts using the multigraded Proj construction of Brenner and Schröer.

The supporting machinery had to be developed. The formalization of the multigraded Proj and multicentered dilatations were established in prior work [9, 10]. The result is a complete, checked proof of a non-trivial theorem in algebraic geometry, with no noetherian, properness, or finite type assumptions on X, and all dependences on the ideals explicit.

This document is organised as follows. Section 1.1 situates the work relative to prior results and existing libraries. Section 1.2 outlines the proof strategy, emphasising the role of the universal property of dilatations. Sections 2–4 contain the proof: the theorem statement, the dilatation and its properties, the reduction to the affine case, and the globalisation. Section 5 lists the correspondence between the statements of this paper and their Lean declarations.

1.1 Relation to other work

The theorem proved below is not new as a statement, and neither is the dilatation on which its proof rests. The dilatation comes from [5], the published version, where multicentered dilatations of rings, schemes and algebraic spaces are introduced, and where the properties used here as Theorem 16 and Lemmas 14 and 15 are established. The multicentered blowup and its universal property, together with the use of the multigraded Proj of Brenner and Schröer [1] to glue the dilatation charts into a scheme, appeared in the fourth arXiv version of that paper [6], but were dropped from the published version of [5]. A second proof, next to a detailed Brenner–Schröer theory, was given in [7]. For $X = \operatorname{Spec} A$ with closed subschemes $Z_1, \dots, Z_k$ cut out by ideals $L_1, \dots, L_k$, the affine blowup is defined as

$$\operatorname{Bl}_Z X \;:=\; \operatorname{Proj}\bigl(\operatorname{Bl}_{L_1,\dots,L_k} A\bigr),$$

the multi-graded Proj of the multi-Rees algebra. It is a final object of the category $\operatorname{Sch}_X^{Z_1,\dots,Z_k\text{-reg}}$ of X-schemes on which all the Z_i become regularly immersed. That is the affine case of Corollary 8, proved below as Theorem 25.

Both ingredients matter, and they enter on an equal footing: the dilatations are what the charts are, and the multi-graded Proj is what makes them into a scheme. For the latter, [6] refers to the multihomogeneous spectra of Brenner and Schröer [1], whose treatment is systematically developed in [7]. In [7] the statement is proved for general X, as [7, Proposition 3.30]: if $T \to X$ is such that every $T \times_X Y_i$ is an effective Cartier divisor on T, then there is a unique morphism of X-schemes $T \to X'$.

The formalisation of the Brenner–Schröer multi-graded Proj construction that the present work relies on throughout was developed by Mayeux and Zhang in [9] (see also the short account [10]), following the algebraic treatment of [7]. That paper formalizes multi-graded Proj schemes and dilatations of rings in Lean 4, as local, ring-level constructions; it stops short of gluing dilatations of quasi-coherent algebras into a global blowup of schemes, a limitation the authors note explicitly. The present work builds on that formalisation without mathematical modification, using it as a black box for everything concerning potions, relevant homogeneous submonoids, and the gluing of Proj from good potion ingredients, and develops the global multicentered blowup and establishes its universal property. Mathlib itself contains no theory of blowups, single- or multicentered, and the multicentered blowup is constructed entirely from these primitives and scheme-theoretic foundations in mathlib.

The construction synthesizes the multi-graded Proj and dilatations as complementary components: dilatations define the affine charts, and the multi-graded Proj provides the gluing machinery. The affine case identifies the charts $D_+(f)$ of $\operatorname{Bl}_Z X$

with dilatations $A[L_1/f, \dots, L_k/f]$, then lifts the local universal property of dilatations to the global setting via the Proj construction. Sections 3 and 4.1 to 4.3 develop this argument in the form the formalisation uses.

The present account provides the globalisation of the universal property and its complete proof. The proof of Theorem 6 is organised so that every step beyond the affine case, namely the base change along open immersions, the comparison isomorphisms on overlaps, the triple overlap maps, the cocycle identity, and the independence of the chosen presentation, is deduced from the affine universal property by uniqueness. The bookkeeping of covers and defining ideals that this requires is managed by the notions of presented centre and centre of Section 2.2. Multi-graded Proj and blowups of this kind are used elsewhere. Tang [13] constructs Gysin maps for regular immersions in a $\mathbf{P}^1$-unstable, non-$\mathbf{A}^1$-invariant theory of motivic spaces and spectra, using pushout-blowups, and observes there that composing several Gysin maps should require the Proj construction for $\mathbb{N}^n$-graded rings, that is an animated version of the theory of [1] and [7].

The formalisation is built on mathlib [4], which supplies schemes, Spec, fibre products, open and closed immersions, gluing data, homogeneous localisation and the theory of non-zero-divisors. The general scheme theory in mathlib, on which everything here depends, was developed by Buzzard et al. [2]. Building on this, Zhang [8] formalized the monograded Proj construction, providing the first non-affine scheme formalization in a theorem prover.

Beyond this general infrastructure, mathlib does not yet contain blowups, dilatations, multigraded Proj, or effective Cartier ideal sheaves. It defines closed immersions, quasi-coherent ideal sheaves (`IdealSheafData`), and the correspondence between them (`overEquivIdealSheafData`). We present a centre by ideals of the rings of an affine cover; Subsection 2.5 shows that this presentation agrees with mathlib's ideal sheaves. We recall the classical statement.

Lemma 1 ([12, Tag 01QN])**.** *Let $i : Z \to X$ be a morphism of schemes. The following are equivalent.*

1. *i is a closed immersion.*
2. *For every affine open $U = \operatorname{Spec} R \subseteq X$ there is an ideal $I \subseteq R$ with $i^{-1}(U) \cong \operatorname{Spec}(R/I)$ as schemes over U.*
3. *There is an affine open cover $X = \bigcup_{j \in J} U_j$, with $U_j = \operatorname{Spec} R_j$, and for each j an ideal $I_j \subseteq R_j$ with $i^{-1}(U_j) \cong \operatorname{Spec}(R_j/I_j)$ as schemes over U_j.*
4. *i induces a homeomorphism of Z onto a closed subset of X and $i^\sharp : \mathcal{O}_X \to i_*\mathcal{O}_Z$ is surjective.*
5. *i induces a homeomorphism of Z onto a closed subset of X, $i^\sharp$ is surjective, and $\ker(i^\sharp) \subseteq \mathcal{O}_X$ is a quasi-coherent sheaf of ideals.*
6. *i induces a homeomorphism of Z onto a closed subset of X, $i^\sharp$ is surjective, and $\ker(i^\sharp) \subseteq \mathcal{O}_X$ is a sheaf of ideals which is locally generated by sections.*

Condition (3) is the one we use. Definition 2 below is that condition with the cover and the ideals promoted from existentially quantified objects to data: rather than asserting that suitable U_j and I_j exist, a presented centre carries a specific choice of

them. This is what makes the Cartier condition of Definition 4 expressible, since that condition is a statement about the chosen ideals, and it is why the equivalence relation of Definition 3 has to be imposed afterwards to forget the choice again.

In the formalisation, a presented centre supplies the data of (3), with charts $\operatorname{Spec} R_j \to X$. It is proved that each $Z_i \to X$ is then a closed immersion, and that its kernel restricts on each chart to I_j (Subsection 2.5).

1.2 Strategy of the proof

Roughly, the proof covers X by affine charts, treats each chart, and glues. On a chart, the main input is the universal property of the multicentered dilatation; the gluing relies mostly on the uniqueness part of the affine universal property.

The proof does not follow [7] closely, nor [6]. The argument there uses multicentered global dilatations of schemes and rests on a body of supporting theory: multi-graded Proj and its charts, quasi-coherent sheaves, closed subschemes, Cartier divisors and their behaviour under base change. Most of that is not available in mathlib, as recorded in Section 1.1. The present proof operates at the ring level, avoiding the need for global scheme-theoretic infrastructure, and constructs the global blowup via covers, charts, gluing data, and uniqueness arguments. Following [7] literally would have meant developing both the global dilatations of schemes and all of their supporting theory. The approach taken here achieves efficiency by building only what is essential for the proof. The multigraded Proj construction of Brenner and Schröer [1] is used to define the blowup by gluing dilatation charts. The universal property of the multicentered dilatation of a ring is then used to prove the universal property of the projective blowup locally, chart by chart. Every global step beyond that, including the gluing and comparison isomorphisms, uses the uniqueness part of the local universal property of the multicentered dilatation.

The shape of the argument is the following.

Existence, locally. In the affine local situation the existence half of the universal property of dilatations applies. Its hypothesis is the Cartier condition with prescribed generators: the images of the a_j are non-zero-divisors and generate the images of the large ideals. So the required morphism into a chart is produced by Theorem 16.

Gluing, by uniqueness. The local morphisms are then glued using the uniqueness part of the same universal property: two local solutions solve the same problem on an overlap, hence agree there. The same argument gives the remaining steps: the comparison isomorphisms $t_{\gamma\delta}$ on double overlaps, the triple overlap maps $t'_{\gamma\delta\epsilon}$, the cocycle identity, the base change isomorphism of Proposition 26, and the independence of the chosen presentation. In each case two objects are shown to solve one problem, and are therefore canonically identified.

Sections 3 to 4 carry this out.

Formal artifact.

The Lean code accompanying this paper is publicly available at https://github.com/rndmx/MULTI-PROJ-BLOW [11], version `v2.1`. It builds with Lean `v4.22.0-rc3` and mathlib at commit `7b332d3`; running `lake exe cache get` followed by `lake build` checks all proofs. Every code excerpt in this paper is preceded by its file and line

numbers in that version. The repository includes portions of the code of [9], updated to this version of mathlib, so that it builds on its own.

2 The theorem

2.1 Centres

Throughout, X is a scheme. The object we blow up is a family of closed subschemes, together with an affine cover on whose charts each member is cut out by an ideal. The definitions allow an arbitrary index type; finiteness is assumed in the theorems.

Definition 2. A *presented centre* on X consists of:

1. a type ι of indices;
2. for each $i \in \iota$, a scheme Z_i together with a morphism $Z_i \to X$;
3. an affine open cover $\mathcal{U} = \big(u_\gamma : U_\gamma := \operatorname{Spec} A_\gamma \to X\big)_{\gamma \in J}$ of X by open immersions;
4. for each i and γ, an ideal $I_{i\gamma} \subseteq A_\gamma$;
5. for each i and γ, an isomorphism of $\operatorname{Spec} A_\gamma$-schemes

$$\operatorname{Spec}\big(A_\gamma / I_{i\gamma}\big) \xrightarrow{\sim} Z_i \times_X \operatorname{Spec} A_\gamma.$$

Condition (5) does two things at once. It forces $Z_i \times_X U_\gamma$ to be affine, and it says that $I_{i\gamma}$ is the ideal cutting it out. So Z_i is a closed subscheme of X, presented on each chart. In Lean this is a single structure.

`Project/Blowups/PreClosAndClos.lean, lines 52--64`

```
variable (X) in
structure PreClos where
  (indnumb : Type)
  (subscheme : indnumb →Scheme)
  [over : ∀(i : indnumb), Scheme.Over (subscheme i) X]
  cov : Scheme.AffineCover.{u+1} (P := @IsOpenImmersion) X
  ideal : ∀(_ : indnumb) (γ : cov.J), Ideal (cov.obj γ)
  condiso : ∀(i : indnumb) (γ : cov.J),
    Spec (CommRingCat.of (cov.obj γ⧸ideal i γ)) ≅
    pullback (f := subscheme i ↘X) (g := cov.map γ)
  condover : ∀(i : indnumb) (γ : cov.J),
    Scheme.Hom.IsOver (condiso i γ).hom
      (Spec (CommRingCat.of (cov.obj γ)))
```

The cover is auxiliary. Two presented centres should be regarded as the same centre when they cut out the same subschemes, whatever charts they use to do it. This is the relation we quotient by.

Definition 3. Two presented centres Z, Z' are *equivalent* if there is a bijection $\sigma : \iota \to \iota'$ of index types and isomorphisms $Z_i \cong Z'_{\sigma(i)}$ over X for every i. A *centre* on X is an equivalence class. We write $\operatorname{Clos}(X)$ for the set of centres.

`Project/Blowups/PreClosAndClos.lean, lines 73--76, 111, 123--124`

```
structure relStructure (Z Z' : PreClos X) where
  indnumb_equiv : Z.indnumb ≃Z'.indnumb
```

```
  subscheme_iso : ∀i, Z.subscheme i ≅Z'.subscheme (indnumb_equiv i)
  subscheme_iso_over : ∀i, Scheme.Hom.IsOver (subscheme_iso i).hom X

def rel : PreClos X →PreClos X →Prop := fun Z Z' => Nonempty (relStructure Z Z')

variable (X)
def Clos := Quotient (relSetoid X)
```

The covers play no role in the relation. That is the whole point: a centre is a family of closed subschemes of X, and the chart data used to present it is forgotten. Everything we build will be built from a presentation, and a substantial part of the work consists in showing that the result does not depend on the presentation.

2.2 Pullback of centres

Centres pull back along an arbitrary morphism, and the construction is the obvious one, but it is worth writing out because the choice of cover on the target is forced and reappears later.

Let $f : X' \to X$ be a morphism and Z a presented centre on X with cover $(u_\gamma : \operatorname{Spec} A_\gamma \to X)_{\gamma \in J}$. The pulled back centre f^*Z has:

- the same index type ι, with subschemes $Z_i \times_X X'$;
- as cover, the family indexed by pairs (γ, β) where $\gamma \in J$ and β runs over an affine open cover of $X' \times_X U_\gamma$, with chart map

$$\operatorname{Spec} A_{\gamma\beta} \longrightarrow X' \times_X U_\gamma \xrightarrow{\mathrm{pr}_1} X';$$

- as ideal at $(i, (\gamma, \beta))$, the image $I_{i\gamma} \cdot A_{\gamma\beta}$ under the induced ring map $A_\gamma \to A_{\gamma\beta}$.

The point is that one cannot simply pull back the cover: $X' \times_X U_\gamma$ is open in X' but need not be affine, so it has to be covered again. This double indexing is why the index type of a pulled back centre is a sigma type, and it is the reason several later statements quantify over pairs.

That f^* respects the equivalence relation of Definition 3 is straightforward, since the relation only involves the subschemes Z_i and base change of schemes preserves isomorphism over the base. So f^* descends to a map

$$f^* : \operatorname{Clos}(X) \longrightarrow \operatorname{Clos}(X').$$

Functoriality, $(g \circ f)^* = f^* \circ g^*$, holds at the level of Clos though not on the nose at the level of presentations, because the chosen affine covers differ. This is a first instance of a phenomenon that recurs throughout: statements that are false for presentations become true for centres.

`Project/Blowups/PreClosAndClos.lean, lines 351--357, 389--390`

```
def pullback_PreClos (X' : Scheme) (f : X' ⟶X) (Z : PreClos X) : PreClos X' where
  indnumb := Z.indnumb
  subscheme i := pullback (Z.subscheme i ↘X) f
  over i := ⟨pullback.snd _ _⟩
  cov := pull_cov X Z X' f
  ideal i γβ:= pull_ideal X Z X' f γβi
  condiso i γβ:= pullback_PreClos_condiso _ γβi
```

```
def pullback_Clos {X' : Scheme} (f : X' ⟶X) : Clos X →Clos X' :=
  Quotient.map (pullback_PreClos X X' f) <| fun Z Z' e => Nonempty.map (pullback_lem X Z Z' X'
      f) e
```

2.3 Cartier centres

The following condition is stated on the charts of a presentation. For a centre given by a finite family of ideal sheaves, it is equivalent to each ideal sheaf being effective Cartier (Proposition 10).

Definition 4. A presented centre Z is *Cartier* if for every $i \in \iota$ and every $\gamma \in J$ the ideal $I_{i\gamma} \subseteq A_\gamma$ is principal and generated by a non-zero-divisor. A centre $\mathcal{Z} \in \mathrm{Clos}(X)$ is Cartier if *some* presentation of $\mathcal{Z}$ is Cartier.

`Project/Blowups/PreClosAndClos.lean, lines 126--146`

```
structure PrePri extends PreClos X where
  [prin : ∀i γ, ideal i γ|>.IsPrincipal]

structure IsPrePri (Z : PreClos X) : Prop where
  prin : ∀i γ, Z.ideal i γ|>.IsPrincipal

attribute [instance] PrePri.prin

structure PreCars extends PrePri X where
  nonzerodiv : ∀i γ, Submodule.IsPrincipal.generator (ideal i γ) ∈nonZeroDivisors (cov.obj γ)

structure IsPreCars (Z : PreClos X) extends IsPrePri _ Z where
  nonzerodiv : ∀i γ, Submodule.IsPrincipal.generator (Z.ideal i γ) ∈nonZeroDivisors (Z.cov.obj
      γ)

structure IsPri (Z : Clos X) : Prop where
  exists_rep : ∃(Z' : PreClos X), IsPrePri _ Z' ∧Quotient.mk'' Z' = Z

def Pri : Set (Clos X) := {x : Clos X | IsPri _ x}

structure IsCars (Z : Clos X) : Prop where
  exists_rep : ∃(Z' : PreClos X), IsPreCars _ Z' ∧Quotient.mk'' Z' = Z
```

Two features of Definition 4 deserve emphasis, because they are what make the proof work and they are easy to blur in informal writing.

First, the existential quantifier sits outside. A centre is Cartier when there is *one* presentation on which all the ideals are principal on all charts. It is not required that every presentation have this property, and indeed most do not.

Second, the generator is required to be a non-zero-divisor. Together with principality this says exactly that each Z_i pulls back to an effective Cartier divisor.

2.4 Statement

Fix a presented centre Z on X with ι finite. We will construct a scheme $\mathrm{Bl}_Z X$ over X. The theorem characterising it is the following.

Theorem 5 (Universal property, presented form)**.** *Let Z be a presented centre on X with finite index type. Let T be a scheme over X, with structure morphism $f : T \to X$. If the pulled back presented centre f^*Z is Cartier, then there is exactly one morphism*

$$\varphi : T \longrightarrow \mathrm{Bl}_Z X$$

over X.

In Lean, with $\mathrm{Bl}_Z X$ written `BlGlob Z`:
`Project/Blowups/BlowupExists.lean, lines 3009--3016`

```
lemma PreProjBlowup_UnivProp
    (Z : PreClos X)
    [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))]
    {T : Scheme} [T.Over X]
    (cond : IsPreCars _ <| pullback_PreClos _ _ (T ↘X) Z) :
    ∃! φ: T ⟶BlGlob Z, Scheme.Hom.IsOver φX := by
```

For $\mathcal{Z} \in \mathrm{Clos}(X)$ with presentation Z, set $\mathrm{Bl}_{\mathcal{Z}} X := \mathrm{Bl}_Z X$ (well defined by Proposition 29).

Theorem 6 (Universal property)**.** *Let $\mathcal{Z} \in \mathrm{Clos}(X)$ be a centre with finite index type and let T be a scheme over X with structure morphism f. If $f^*\mathcal{Z}$ is Cartier, then there is exactly one morphism $T \to \mathrm{Bl}_{\mathcal{Z}} X$ over X.*

`Project/Blowups/BlowupExists.lean, lines 3392--3396`

```
theorem GlobalBlowup_UnivProp (Z : Clos X) [DecidableEq Z.out.indnumb] [Fintype Z.out.indnumb]
    [(i : Z.out.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.out.indnumb))]
    {T : Scheme} [T.Over X]
    (cond : IsCars _ <| pullback_Clos (T ↘X) Z) :
    ∃! φ: T ⟶GlobalBlowup Z, Scheme.Hom.IsOver φX := by
```

The instances `Fintype` and `DecidableEq` are used by the construction of $\mathrm{Bl}_Z X$ (Section 3). Since `Fintype` is a subsingleton, the result does not depend on them; Corollary 11 assumes only `Finite`.

Alongside these sits the statement that the blowup itself has the property in question.

Theorem 7. *The centre pulled back to $\mathrm{Bl}_Z X$ is Cartier.*

`Project/Blowups/BlowupExists.lean, lines 3342--3345`

```
lemma BlGlob_IsCars (Z : PreClos X)
    [DecidableEq Z.indnumb] [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    IsCars _ (pullback_Clos ((BlGlob Z) ↘X) (Quotient.mk'' Z)) := by
```

The scheme $\mathrm{Bl}_{\mathcal{Z}} X$ itself, for a centre rather than a presentation, is obtained by choosing the representative that the quotient supplies.
`Project/Blowups/BlowupExists.lean, lines 3382--3384`

```
noncomputable def GlobalBlowup (Z : Clos X) [DecidableEq Z.out.indnumb] [Fintype Z.out.indnumb]
    [(i : Z.out.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.out.indnumb))] : Scheme :=
  BlGlob Z.out
```

Combining Theorems 6 and 7 gives the following. Let $\mathcal{C}_{\mathcal{Z}}$ be the category whose objects are X-schemes T such that $f^*\mathcal{Z}$ is Cartier, and whose morphisms are morphisms over X. Then:

Corollary 8. $\mathrm{Bl}_{\mathcal{Z}} X$ *is a terminal object of* $\mathcal{C}_{\mathcal{Z}}$.

No assumption is made on X or T.

Remark 9. For ι a singleton, this is [12, Tag 0806].

2.5 Formulation with mathlib ideal sheaves

For an ideal sheaf $\mathcal{I}$ on X, mathlib provides a closed subscheme $V(\mathcal{I}) \to X$ (`IdealSheafData.subscheme`); closed immersions into X correspond to ideal sheaves on X (`overEquivIdealSheafData`). For $f : T \to X$, we write $f^{-1}\mathcal{I}$ for the ideal sheaf of $T \times_X V(\mathcal{I}) \to T$ (`IdealSheafData.comap`).

From ideal sheaves to centres. Let $(\mathcal{I}_i)_{i\in\iota}$ be a family of ideal sheaves on X. It defines a presented centre (`toPreClos`): the schemes are the $V(\mathcal{I}_i)$, the cover consists of the charts $\operatorname{Spec}\mathcal{O}_X(U) \to X$ for all affine opens U, the ideals are the $\mathcal{I}_i(U)$, and the isomorphisms of item (5) are `specQuotientIsoPullback`. We write $[\mathcal{I}] \in \mathrm{Clos}(X)$ for its class (`toClos`).

From centres to ideal sheaves. Let Z be a presented centre. Each $Z_i \to X$ is a closed immersion: this is local on the target, and over the chart $\operatorname{Spec} A_\gamma$ the base change $Z_i \times_X \operatorname{Spec} A_\gamma \to \operatorname{Spec} A_\gamma$ is identified with $\operatorname{Spec}(A_\gamma/I_{i\gamma}) \to \operatorname{Spec} A_\gamma$ (`PreClos.isClosedImmersion`). On $u_\gamma(U_\gamma)$, the kernel of $Z_i \to X$ corresponds to $I_{i\gamma}$ under $\mathcal{O}_X(u_\gamma(U_\gamma)) \cong A_\gamma$ (`PreClos.ker_ideal_chart`). Moreover, $[Z] = [\ker(Z_i \to X)]_i$ in $\mathrm{Clos}(X)$ (`PreClos.mk_eq_toClos_ker`). A presented centre adds the choice of an affine cover, on which the Cartier condition of Definition 4 is stated; $\mathrm{Clos}(X)$ forgets this choice.

Cartier conditions. An ideal sheaf $\mathcal{I}$ on X is *effective Cartier* if every point of X has an affine open neighbourhood U such that $\mathcal{I}(U)$ is generated by a non-zero-divisor of $\mathcal{O}_X(U)$.

`Project/Blowups/IdealSheaf.lean, lines 203--209`

```
def IsPrincipalNZDOn (I : X.IdealSheafData) (U : X.affineOpens) : Prop :=
  ∃f ∈nonZeroDivisors Γ(X, U), I.ideal U = Ideal.span {f}

def IsEffectiveCartier (I : X.IdealSheafData) : Prop :=
  ∀x : X, ∃U : X.affineOpens, x ∈U.1 ∧I.IsPrincipalNZDOn U
```

Proposition 10. *Let $(\mathcal{I}_i)_{i\in\iota}$ be a finite family of ideal sheaves on X and $f : T \to X$ a morphism. Then $f^*[\mathcal{I}]$ is Cartier if and only if each $f^{-1}\mathcal{I}_i$ is effective Cartier (`isCars_pullback_toClos_iff`).*

Proof Suppose each $f^{-1}\mathcal{I}_i$ is effective Cartier. Generation by a non-zero-divisor passes to basic opens, and two affine neighbourhoods of a point contain a common basic open neighbourhood. Since ι is finite, induction on ι shows that every point of T has an affine open neighbourhood on which all the $f^{-1}\mathcal{I}_i$ are generated by non-zero-divisors (`exists_isPrincipalNZDOn`). Let S be the set of such affine opens. Presenting $(f^{-1}\mathcal{I}_i)_i$ on the cover by S gives a Cartier presented centre (`toPreClosOn`, `isPreCars_of_generators`). It is equivalent to $f^*[\mathcal{I}]$, through the isomorphisms $V(f^{-1}\mathcal{I}_i) \cong T \times_X V(\mathcal{I}_i)$ over T.

Conversely, let W be a Cartier presented centre equivalent to $f^*[\mathcal{I}]$, with bijection σ of index sets. The kernel of $W_k \to T$ is $f^{-1}\mathcal{I}_{\sigma(k)}$. For $x \in T$, take a chart of W whose image contains x. By `PreClos.ker_ideal_chart`, the sections of $f^{-1}\mathcal{I}_{\sigma(k)}$ on that image correspond to the ideal of W on the chart, which is generated by a non-zero-divisor. □

Corollary 11. *Let $(\mathcal{I}_i)_{i\in\iota}$ be a finite family of ideal sheaves on X. There is a scheme B over X such that each $\mathcal{I}_i$ pulls back to an effective Cartier ideal sheaf on B, and every X-scheme with this property has a unique morphism to B over X.*

`Project/Blowups/IdealSheaf.lean, lines 358--361`

```
theorem exists_blowup_isEffectiveCartier [Finite ι] :
    ∃(B : Scheme) (_ : B.Over X), (∀ i, ((I i).comap (B ↘X)).IsEffectiveCartier) ∧
      ∀(T : Scheme) [T.Over X], (∀ i, ((I i).comap (T ↘X)).IsEffectiveCartier) →
        ∃! φ: T ⟶B, φ.IsOver X := by
```

Proof Apply Theorem 6 to $[\mathcal{I}]$ and use Proposition 10. □

For a finite family of closed immersions $Z_i \to X$, the same holds with $\mathcal{I}_i = \ker(Z_i \to X)$ (`exists_blowup_isEffectiveCartier_of_isClosedImmersion`).

3 Dilatation

This section contains the construction that drives everything, following [5] and [9]. Some of the results on dilatations here are new to this project.

3.1 Multicenters

Definition 12. Let A be a commutative ring. A *multicenter* on A is a triple $F = (\Lambda, (I_j)_{j\in\Lambda}, (a_j)_{j\in\Lambda})$ where Λ is a type, each I_j is an ideal of A and each a_j is an element of A. The *large ideal* at j is

$$\mathcal{L}_j \;:=\; I_j + (a_j).$$

`Project/Dilatation/Multicenter.lean, lines 19--23, 39`

```
@[ext]
structure Multicenter : Type (u+1) where
  (index : Type)
  (ideal : index →Ideal A)
  (elem : index →A)

def LargeIdeal (i : F.index) : Ideal A := F.ideal i + Ideal.span {F.elem i}
```

The constructions below depend on F only through the pairs $(\mathcal{L}_j, a_j)$; I_j enters only via $\mathcal{L}_j$. Ideals $\mathcal{L}_j$ with elements $a_j \in \mathcal{L}_j$, for $j \in \Lambda$, form a multicenter (take $I_j = \mathcal{L}_j$), whose large ideals are the $\mathcal{L}_j$. The form (I_j, a_j) is the one of [5] and of the formalisation [9].

For $\nu \in \Lambda^{(\mathbb{N})}$, a finitely supported function $\Lambda \to \mathbb{N}$, write

$$\mathcal{L}^\nu := \prod_j \mathcal{L}_j^{\nu_j}, \qquad a^\nu := \prod_j a_j^{\nu_j},$$

both finite products.

3.2 The dilatation algebra

Definition 13. The *dilatation* of A along F is

$$A[F] \;:=\; \left\{ \tfrac{m}{a^\nu} \;:\; \nu \in \Lambda^{(\mathbb{N})},\ m \in \mathcal{L}^\nu \right\} \Big/ \sim,$$

where

$$\frac{m}{a^\nu} \sim \frac{n}{a^\mu} \iff \exists\, \beta \in \Lambda^{(\mathbb{N})} : \quad m\, a^{\beta+\mu} \;=\; n\, a^{\beta+\nu}.$$

Addition and multiplication are the obvious ones,

$$\frac{m}{a^\nu} + \frac{n}{a^\mu} = \frac{m a^\mu + n a^\nu}{a^{\nu+\mu}}, \qquad \frac{m}{a^\nu} \cdot \frac{n}{a^\mu} = \frac{mn}{a^{\nu+\mu}},$$

and one checks these are well defined. Note that $m a^\mu \in \mathcal{L}^{\nu+\mu}$ because $a_j \in \mathcal{L}_j$, which is where the definition of the large ideal is used. The map $A \to A[F]$, $x \mapsto x/a^0$, makes $A[F]$ an A-algebra.

`Project/Dilatation/Multicenter.lean, lines 50--56, 89--90`

```
structure PreDil where
  pow : F.index →₀ ℕ
  num : A
  num_mem : num ∈ F.LargeIdeal ^ pow

def r : F.PreDil → F.PreDil → Prop := fun x y =>
  ∃ β: F.index →₀ ℕ, x.num * F.elem^(β + y.pow) = y.num * F.elem^(β + x.pow)

variable (F) in
def Dilatation := Quotient F.setoid
```

Informally, $A[F] = A\big[\mathcal{L}_j/a_j : j \in \Lambda\big]$, the subring of the total localisation generated by the fractions ℓ/a_j with $\ell \in \mathcal{L}_j$. The quotient presentation above avoids having to embed anything into a localisation.

3.3 The two Cartier lemmas

The following pair of statements is the reason dilatations are the right local model. They say that in $A[F]$ the large ideals become principal, generated by non-zero-divisors.

Lemma 14. *For every ν, the image of a^ν in $A[F]$ is a non-zero-divisor.*

Lemma 15. *For every ν, inside $A[F]$ one has*

$$\big(a^\nu\big) \;=\; \mathcal{L}^\nu \cdot A[F].$$

`Project/Dilatation/Multicenter.lean, lines 374, 386--388`

```
lemma nonzerodiv_image (v : F.index →₀ ℕ) : algebraMap A A[F] (F.elem^v) ∈nonZeroDivisors A[F]
     := by

lemma image_elem_LargeIdeal_equal (v : F.index →₀ ℕ) :
 Ideal.span ({algebraMap A A[F] (F.elem^v)}) =
    Ideal.map (algebraMap A A[F]) (F.LargeIdeal^v) := by
```

The proof of Lemma 14 is short and shows what the relation is doing. Suppose $\frac{m}{a^\mu} \cdot a^\nu = 0$ in $A[F]$. By definition of $\sim$ there is β with $m\,a^\nu a^\beta = 0$ in A. But then taking $\beta' = \nu + \beta$ witnesses $\frac{m}{a^\mu} = 0$. So the element we inverted is a non-zero-divisor not because it was one in A, which it need not be, but because the equivalence relation was set up to kill exactly the elements it annihilates.

Lemma 15 is the inclusion $\mathcal{L}^\nu A[F] \subseteq (a^\nu)$, which holds since $m \in \mathcal{L}^\nu$ gives $m = a^\nu \cdot \frac{m}{a^\nu}$, together with the reverse inclusion, which holds since $a^\nu \in \mathcal{L}^\nu$.

3.4 The universal property

Theorem 16. *Let B be an A-algebra such that*

1. *the image of a_j in B is a non-zero-divisor for every j, and*
2. *$(a_j) = \mathcal{L}_j B$ inside B, for every j.*

Then there is exactly one morphism of A-algebras $A[F] \to B$.

`Project/Dilatation/Multicenter.lean, lines 545--548, 622--625`

```
def desc [Algebra A B]
    (non_zero_divisor : ∀i : F.index, (algebraMap A B) (F.elem i) ∈nonZeroDivisors B)
    (gen : ∀i, Ideal.span {(algebraMap A B) (F.elem i)} = Ideal.map (algebraMap A B)
         (F.LargeIdeal i)) :
    A[F] →ₐ[A] B where

lemma lemma_exists_unique_morphism [Algebra A B]
    (non_zero_divisor : ∀i : F.index, (algebraMap A B) (F.elem i) ∈nonZeroDivisors B)
    (gen : ∀i, Ideal.span {(algebraMap A B) (F.elem i)} = Ideal.map (algebraMap A B)
         (F.LargeIdeal i))
    (χ' : A[F] →ₐ[A] B) : χ' = desc F non_zero_divisor gen := by
```

Existence and uniqueness both come from the same observation. Given ν and $m \in \mathcal{L}^\nu$, hypothesis (2) gives $m \in \mathcal{L}^\nu B = (a^\nu)$, so there is $b \in B$ with $m = a^\nu b$, and hypothesis (1) says this b is unique. Send $\frac{m}{a^\nu}$ to that b. Any A-algebra map χ must satisfy $a^\nu \chi\big(\frac{m}{a^\nu}\big) = \chi(m) = m$, so it must send $\frac{m}{a^\nu}$ to the same b. Existence and uniqueness are the two halves of "divide by a^ν, and the quotient is unique".

The hypotheses on B are the Cartier condition with prescribed generators. So Theorem 16 reads:

$A[F]$ is the initial A-algebra B such that, for every j, $\mathcal{L}_j B = (a_j)$ with a_j a non-zero-divisor in B.

That is the affine case of Corollary 8, in ring-theoretic form. Everything in Section 4 is an elaboration of this one statement: first upgrading it from a single multicenter to a family of ideals with all of its presentations, then from affine to global.

The following special case follows from Theorem 16 by uniqueness.

Corollary 17. *If every a_j is already a non-zero-divisor in A and $\mathcal{L}_j = (a_j)$, then $A \to A[F]$ is an isomorphism.*

Indeed $B = A$ then satisfies the hypotheses of Theorem 16, so there is a unique $A[F] \to A$, and both composites with $A \to A[F]$ are identities by the uniqueness clause applied to A and to $A[F]$ in turn.

`Project/Dilatation/Multicenter.lean, lines 657--658`

```
def ofFamilyIso {index : Type} {c : index →A} (c0 : ∀i, c i ∈nonZeroDivisors A) :
    A[ofFamily index c] ≃ₐ[A] A :=
```

3.5 Functoriality

Let $A \to B$ be a ring map and F a multicenter on A. Pushing forward the ideals and the elements gives a multicenter F_B on B with the same index type, ideals $I_j B$ and elements the images of a_j. One checks $\mathcal{L}_j(F_B) = \mathcal{L}_j(F)B$, so the large ideals also push forward.

Proposition 18. *There is a unique morphism of A-algebras $A[F] \to B[F_B]$ compatible with $A \to B$.*

Again this is Theorem 16: in $B[F_B]$ the element a_j is a non-zero-divisor generating $\mathcal{L}_j(F)B[F_B]$, by Lemmas 14 and 15 applied over B. So $A[F] \to B[F_B]$ exists and is unique. The pushed forward multicenter F_B is `image_mult`, and the map is built by `desc`, that is by the universal property.

`Project/Dilatation/Multicenter.lean, lines 772--775, 784`

```
def image_mult [Algebra A B] : Multicenter B :=
  { index := _
    ideal i := Ideal.map (algebraMap A B) (F.ideal i)
    elem i := algebraMap A B (F.elem i)}

def functo_dila_alg [Algebra A B] : A[F] →ₐ[A] B[image_mult (B := B) F] :=
  desc F ...
```

When $A \to B$ is flat one can say more: the natural map

$$B \otimes_A A[F] \longrightarrow B[F_B]$$

is an isomorphism. Flatness is genuinely needed here, since the construction of the inverse uses that a non-zero-divisor stays a non-zero-divisor after a flat base change.

3.6 Presentations

We now fix a finite index type ι and a family $L = (L_i)_{i\in\iota}$ of ideals of A. This is the affine centre we want to blow up. A single multicenter is not enough, because a multicenter comes with a choice of the elements a_j, and there is no canonical choice. The construction takes all choices at once.

Definition 19. A *presentation* of L is a multicenter F on A with finite index type Λ, together with maps $\Psi : \Lambda \to \iota$ and $s : \iota \to \Lambda$ such that

$$\Psi \circ s = \mathrm{id}_\iota, \qquad \mathcal{L}_j = L_{\Psi(j)} \quad \text{for all } j \in \Lambda.$$

We write $\mathcal{M}(L)$ for the collection of presentations.

`Project/Blowups/Bl.lean, lines 18--24`

```
structure Mu : Type (u + 1) where
multicenter : Multicenter A
[fin : Fintype multicenter.index]
Ψ: multicenter.index →ι
sec : ι→multicenter.index
surj : ∀i, Ψ(sec i) = i
cond : ∀i, multicenter.LargeIdeal i = L (Ψ i)
```

Unwinding, a presentation is a finite set Λ of "chart directions", each labelled by the ideal $L_{\Psi(j)}$ it refers to, and for each j an element $a_j \in L_{\Psi(j)}$, presented as

$$L_{\Psi(j)} \;=\; I_j + (a_j).$$

The section s exists if and only if Ψ is surjective. Nothing requires Ψ to be injective, so an ideal may appear with several different chosen elements.

3.7 Dilatations as charts of a multigraded Proj

The dilatations $A[F]$, $F \in \mathcal{M}(L)$, are the charts of a single scheme. To glue them we realise each as an affine piece of a multigraded Proj.

Let

$$\mathrm{Rees}(L) \;:=\; \bigoplus_{v\in\iota^{(\mathbb{N})}} L^v, \qquad L^v := \prod_i L_i^{v_i},$$

the multi-Rees algebra of the family, graded by $\mathbb{N}^\iota$ and then by $\mathbb{Z}^\iota$ through the inclusion $\mathbb{N}^\iota \hookrightarrow \mathbb{Z}^\iota$. For a homogeneous submonoid S of $\mathrm{Rees}(L)$, let

$$\mathrm{Pot}(S) \;:=\; \left(S^{-1}\,\mathrm{Rees}(L)\right)_0$$

be the degree zero part of the homogeneous localisation, called the *potion* of S in the formalisation. The multigraded Proj is glued from the affine schemes $\operatorname{Spec}\mathrm{Pot}(S)$ as S runs over the relevant finitely generated homogeneous submonoids. Here the *saturation* $\bar{S}$ of S is the submonoid of homogeneous elements of $\mathrm{Rees}(L)$ dividing some element of

S, and S is *relevant* if for every $v \in \mathbb{Z}^\iota$ some positive multiple nv lies in the subgroup generated by the degrees of $\bar{S}$.

Given a presentation $F \in \mathcal{M}(L)$, let $S_F \subseteq \operatorname{Rees}(L)$ be the submonoid generated by the elements

$$a_j \in L_{\Psi(j)} = \big(\operatorname{Rees} L\big)_{e_{\Psi(j)}}, \qquad j \in \Lambda,$$

that is, a_j placed in degree $e_{\Psi(j)}$, the $\Psi(j)$-th basis vector.

Proposition 20. *There is a canonical isomorphism of A-algebras*

$$A[F] \xrightarrow{\sim} \operatorname{Pot}(S_F).$$

A homogeneous element of S_F of degree $\nu \in \mathbb{N}^\iota$ is a product a^λ for $\lambda \in \Lambda^{(\mathbb{N})}$ with $\sum_j \lambda_j e_{\Psi(j)} = \nu$. A degree zero element of the localisation $S_F^{-1} \operatorname{Rees}(L)$ is therefore a fraction

$$\frac{m}{a^\lambda}, \qquad m \in \operatorname{Rees}(L)_\nu = L^\nu,$$

and the defining condition $\mathcal{L}_j = L_{\Psi(j)}$ of a presentation says exactly that

$$L^\nu \;=\; \prod_i L_i^{\nu_i} \;=\; \prod_j \mathcal{L}_j^{\lambda_j} \;=\; \mathcal{L}^\lambda.$$

So the numerators available on the two sides coincide. The equivalence relation also matches: two fractions in a homogeneous localisation agree when they agree after multiplying by some element of S_F, and elements of S_F are exactly the a^β, which is the relation of Definition 13. The map itself is again given by the universal property of the dilatation, and the isomorphism is obtained from injectivity and surjectivity, proved separately.

`Project/Blowups/Bl.lean, lines 114--116, 271--272, 605--607`

```
def clo_mu (P : Mu L) :
    HomogeneousSubmonoid (ReesAlgebra.intGrading L) :=
  HomogeneousSubmonoid.closure

def clo_mu_mor (P : Mu L) : A[P.multicenter] →ₐ[A] (clo_mu L P).Potion :=
  Multicenter.desc P.multicenter

def Mu_mor_iso (P : Mu L) :
    A[P.multicenter] ≃ₐ[A] (clo_mu L P).Potion :=
  AlgEquiv.ofBijective (clo_mu_mor ..) <| ⟨clo_mu_mor_inj L P, clo_mu_mor_surj L P⟩
```

The inclusion $\mathcal{L}_j \subseteq L_{\Psi(j)}$ is used to define the map, and the reverse inclusion for surjectivity.

For the gluing we also need to know that S_F is an admissible index for the Proj construction, meaning finitely generated and relevant. Finite generation is clear, since Λ is finite and S_F is generated by the a_j. Relevance amounts to the degrees of $\bar{S}_F$ generating a finite-index subgroup of $\mathbb{Z}^\iota$; they generate $\mathbb{Z}^\iota$, since $S_F \subseteq \bar{S}_F$, a_j has degree $e_{\Psi(j)}$, and Ψ is surjective.

Definition 21. $\operatorname{Bl}^{\mathrm{aff}}(L) :=$ Proj of $\operatorname{Rees}(L)$ taken over the family $\{S_F : F \in \mathcal{M}(L)\}$, that is, the scheme obtained by gluing the affine charts $\operatorname{Spec}\operatorname{Pot}(S_F) \cong \operatorname{Spec} A[F]$,

$F \in \mathcal{M}(L)$, along the standard Proj gluing data. It carries a structure morphism $\mathrm{Bl}^{\mathrm{aff}}(L) \to \operatorname{Spec} A$ induced by $A \cong \mathrm{Rees}(L)_0$.

`Project/Blowups/Bl.lean, lines 703--706, 721--722, 726--728`

```
def map_index (P : Mu L) :
    GoodPotionIngredient (ReesAlgebra.intGrading L) where
  toHomogeneousSubmonoid := clo_mu L P
  relevant := clo_mu_rel L P

def BlMu : Scheme :=
  Proj (τ := Mu L) (map_index L)

instance BlMuOverSpec : Scheme.Over (BlMu L) (Spec <| CommRingCat.of A) where
  hom := (GoodPotionIngredient.over (F := map_index L)) ≫
    Spec.map (CommRingCat.ofHom <| ReesAlgebra.degreeZeroIso' L)
```

The field `relevant` is the relevance condition discussed above, and the structure morphism to $\operatorname{Spec} A$ is induced by the isomorphism $A \cong \mathrm{Rees}(L)_0$ recorded as `degreeZeroIso'`.

Two remarks. First, the gluing data is inherited from the ambient multigraded Proj, so the overlaps and the cocycle condition are inherited from the Proj construction, requiring no additional verification. Second, we do not need to know that $\mathrm{Bl}^{\mathrm{aff}}(L)$ is all of $\operatorname{Proj} \mathrm{Rees}(L)$. It is the union of the dilatation charts, and every statement we prove is scoped to those charts, reflecting a deliberate architectural choice that avoids the need for a comparison that would otherwise complicate the proof.

To summarise the picture:

$$\begin{array}{ccccc} \operatorname{Spec} A[F] & \xrightarrow{\sim} & \operatorname{Spec} \operatorname{Pot}(S_F) & \hookrightarrow & \mathrm{Bl}^{\mathrm{aff}}(L) \\ & \searrow & & & \downarrow \\ & & & & \operatorname{Spec} A \end{array}$$

Each presentation gives a dilatation, each dilatation is an affine chart, and the charts glue to the affine blowup.

4 Proof of the theorem

The proof has two halves. First the affine case: $\mathrm{Bl}^{\mathrm{aff}}(L)$ is terminal among affine-based $\operatorname{Spec} A$-schemes on which L becomes Cartier. Then the global case: the local blowups over the charts of X are glued, and the gluing is governed at every stage by the affine uniqueness statement.

4.1 The affine centre is Cartier on $\mathrm{Bl}^{\mathrm{aff}}(L)$

The first thing to prove is that $\mathrm{Bl}^{\mathrm{aff}}(L)$ does what it is supposed to do.

Proposition 22. *The centre L pulled back to $\mathrm{Bl}^{\mathrm{aff}}(L)$ is Cartier.*

The presentation exhibiting this is: take as cover the family of dilatation charts $\operatorname{Spec} A[F]$, $F \in \mathcal{M}(L)$, indexed by $\mathcal{M}(L)$ itself, and as ideal on the chart F the image $L_i \cdot A[F]$.

For each i and each F, we must produce a generator of $L_i \cdot A[F]$ that is a non-zero-divisor. Choose j with $\Psi(j) = i$ (in the formalisation, $j = s(i)$), so $\mathcal{L}_j = L_i$. Then Lemmas 14 and 15, applied with $\nu = e_j$, give

$$L_i \cdot A[F] \;=\; \mathcal{L}_j \cdot A[F] \;=\; \big(a_j\big), \qquad a_j \text{ a non-zero-divisor in } A[F].$$

So the generator is a_j. The formal proof passes these two lemmas to a criterion that checks the Cartier property chart by chart.

`Project/Blowups/Cars.lean, lines 64--68`

```
lemma BlMuPreClos_IsPreCars (A : CommRingCat) (L : ι→Ideal A) :
    IsPreCars _ (BlMuPreClos A L) := by
  refine isPreCars_of_generators (BlMuPreClos A L)
     (fun i (P : Mu L) => (Mu_mor_iso L P).toRingEquiv
       (algebraMap A A[P.multicenter] (P.multicenter.elem (P.sec i)))) ?_ ?_
```

The presented centre and the Cartier statement are as follows. Its cover is `BlMuAffineCover`, the family of dilatation charts, and its ideal on the chart P is the image of L_i.

`Project/Blowups/Cars.lean, lines 43--48, 89--91`

```
def BlMuPreClos (A : CommRingCat) (L : ι→Ideal A) : PreClos (BlMu L) where
  indnumb := ι
  subscheme i := pullback ((loc_to_PreClos A L).subscheme i ↘Spec A) (BlMu L ↘Spec A)
  over i := ⟨pullback.snd _ _⟩
  cov := BlMuAffineCover A L
  ideal i P := Ideal.map (algebraMap A ((map_index L P).Potion)) (L i)

lemma blowups_Cars (A : CommRingCat) (L : ι→Ideal A) :
    IsCars _ <|
      pullback_Clos (BlMu L ↘(Spec (CommRingCat.of A))) (loc_to_Clos A L) :=
```

The term `P.multicenter.elem (P.sec i)` is a_j for $j = s(i)$, and `Mu_mor_iso` is the isomorphism of Proposition 20 transporting it into the potion.

A consequence we use repeatedly: an open subscheme of $\mathrm{Bl}^{\mathrm{aff}}(L)$ still has the property, since open immersions are flat and the Cartier property is stable under flat base change. In the formalisation this is `open_of_blowup_IsCars` (`Project/Blowups/Cars.lean`, lines 100–107).

4.2 Affine uniqueness

Proposition 23. *Let T be a scheme over $\operatorname{Spec} A$ on which L becomes Cartier. Then any two morphisms $T \to \mathrm{Bl}^{\mathrm{aff}}(L)$ over $\operatorname{Spec} A$ are equal.*

Proof Let $p : T \to \operatorname{Spec} A$ be the structure morphism, let $j_F : \operatorname{Spec} A[F] \cong \operatorname{Spec} \mathrm{Pot}(S_F) \to \mathrm{Bl}^{\mathrm{aff}}(L)$, $F \in \mathcal{M}(L)$, be the chart inclusions of Definition 21, and let $\varphi, \varphi' : T \to \mathrm{Bl}^{\mathrm{aff}}(L)$ be morphisms over $\operatorname{Spec} A$. Two morphisms of schemes that agree on an open neighbourhood of every point are equal, so it suffices to fix $x \in T$ and find such a neighbourhood.

By hypothesis and Definition 4, the pulled back centre has a Cartier presentation Z, with an affine open cover $(u_\gamma : \operatorname{Spec} B_\gamma \to T)_\gamma$, and by Definition 3 there is a bijection σ from the index type of Z to ι. Since $p \circ u_\gamma = \operatorname{Spec}(A \to B_\gamma)$, the ideal of Z of index k on B_γ equals $L_{\sigma(k)}B_\gamma$ (`PreClos_ideal_eq_of_rel`). Reindexing by σ, for each $i \in \iota$ and each γ there is a non-zero-divisor $z_{i\gamma} \in B_\gamma$ with $L_i B_\gamma = (z_{i\gamma})$.

Choose γ with $x \in u_\gamma(\operatorname{Spec} B_\gamma)$, and $F, G \in \mathcal{M}(L)$ with $\varphi(x) \in j_F(\operatorname{Spec} A[F])$ and $\varphi'(x) \in j_G(\operatorname{Spec} A[G])$. Choose an affine open $\operatorname{Spec} B \ni x$ contained in $\varphi^{-1}(j_F(\operatorname{Spec} A[F])) \cap \varphi'^{-1}(j_G(\operatorname{Spec} A[G])) \cap u_\gamma(\operatorname{Spec} B_\gamma)$. Since j_F, j_G, u_γ are open immersions, we obtain ring maps $g : A[F] \to B$, $g' : A[G] \to B$, $h : B_\gamma \to B$ with $\varphi|_{\operatorname{Spec} B} = j_F \circ \operatorname{Spec}(g)$ and $\varphi'|_{\operatorname{Spec} B} = j_G \circ \operatorname{Spec}(g')$. As all morphisms are over $\operatorname{Spec} A$, g and g' are A-algebra maps and $A \to B$ factors through h.

The map h is flat, because $\operatorname{Spec} B \to \operatorname{Spec} B_\gamma$ is an open immersion of affine schemes (Section 4.1). Hence $c_i := h(z_{i\gamma})$ is a non-zero-divisor of B and $L_i B = (c_i)$ for every $i \in \iota$.

Let $F \sqcup G \in \mathcal{M}(L)$ be the presentation indexed by $\Lambda_F \sqcup \Lambda_G$, carrying the pairs of F and of G, with section that of F (`union_Mu`), and let $\lambda_F : A[F] \to A[F \sqcup G]$ and $\lambda_G : A[G] \to A[F \sqcup G]$ be the structure maps (`dilationToUnion_left`, `dilationToUnion_right`). Since each c_i is a non-zero-divisor with $L_i B = (c_i)$, there is an A-algebra map $g'' : A[F \sqcup G] \to B$ with $g = g'' \circ \lambda_F$ and $g' = g'' \circ \lambda_G$ (Lemmas 15, 14 and Theorem 16; `lemm_dila_double_union`).

One has $S_F, S_G \leq S_{F \sqcup G}$, and under the isomorphisms of Proposition 20 the localisation maps $\operatorname{Pot}(S_F) \to \operatorname{Pot}(S_{F\sqcup G})$ and $\operatorname{Pot}(S_G) \to \operatorname{Pot}(S_{F\sqcup G})$ are λ_F and λ_G (`Mu_mor_iso_commutes_ringHom'`, `Mu_mor_iso_commutes_ringHom_right'`). The gluing data of $\mathrm{Bl}^{\mathrm{aff}}(L)$ then give $j_F \circ \operatorname{Spec}(\lambda_F) = j_{F\sqcup G} = j_G \circ \operatorname{Spec}(\lambda_G)$ (`proj_glue_condition`). Therefore

$$\varphi|_{\operatorname{Spec} B} = j_F \circ \operatorname{Spec}(\lambda_F) \circ \operatorname{Spec}(g'') = j_{F\sqcup G} \circ \operatorname{Spec}(g'') = j_G \circ \operatorname{Spec}(\lambda_G) \circ \operatorname{Spec}(g'') = \varphi'|_{\operatorname{Spec} B}.$$

□

`Project/Blowups/UniqueBlowup.lean, lines 254--261`

```
theorem ProjBlowup_UnivProp_unicity_affine
  (A : CommRingCat.{u+1}) (L : ι→Ideal A) [fin : Fintype ι]
  {T : Scheme} [T.Over (Spec A)]
  (cond : IsCars _ (pullback_Clos (T ↘Spec A) (loc_to_Clos A L)))

  (φ φ' : T ⟶BlMu L)
  (φ_over : Scheme.Hom.IsOver φ(Spec A))
  (φ'_over : Scheme.Hom.IsOver φ' (Spec A)) : φ= φ' := by
```

4.3 Affine existence

Proposition 24. *Let T be a scheme over $\operatorname{Spec} A$ on which L becomes Cartier. Then there is a morphism $T \to \mathrm{Bl}^{\mathrm{aff}}(L)$ over $\operatorname{Spec} A$.*

Proof We follow `ProjBlowup_UnivProp_existence_affine_preclo` (`Project/Blowups/BlowupExists.lean`). Write $e_i \in \mathbb{Z}^\iota$ for the basis vectors and $d = \sum_i e_i$. For a homogeneous element f of a Rees algebra, $\langle f \rangle$ denotes the submonoid it generates. Saturation and relevance are as in Section 3.7; a homogeneous submonoid and its saturation have the same potion (`equivBarPotion`). If S is generated by finitely many homogeneous elements with product f, then $\bar{S} = \overline{\langle f \rangle}$, since each generator divides f and $f \in S$; hence $\operatorname{Pot}(S) = \operatorname{Pot}\langle f \rangle$.

Step 1 (local data). Choose a Cartier presentation of the pullback of L to T (Definition 4). It gives an affine open cover $(u_\gamma : \operatorname{Spec} B_\gamma \to T)_\gamma$ and non-zero-divisors $c_i \in B_\gamma$ with

$L_iB_\gamma = (c_i)$, after reindexing by the bijection of Definition 3 (`PreClos_ideal_eq_of_rel`). Fix γ and write $B = B_\gamma$; the map $\operatorname{Spec} B \to \operatorname{Spec} A$ is Spec of a ring map $A \to B$.

Step 2 (B as a potion). The pairs (L_iB, c_i), with $\Psi = s = \mathrm{id}$, form a presentation F_B of LB over B, since $\mathcal{L}_i = L_iB + (c_i) = (c_i) = L_iB$. Put $c = \prod_i c_i t^{e_i} \in \operatorname{Rees}(LB)$, of degree d. By Corollary 17, Proposition 20 and the preliminary remark, $B \cong B[F_B] \cong \operatorname{Pot}(S_{F_B}) = \operatorname{Pot}\langle c\rangle$ as B-algebras.

Step 3 (expansion). Since $c_i \in L_iB$, there are a finite set $t_i \subseteq L_i$ and $\lambda_i(a) \in B$ with $c_i = \sum_{a\in t_i} \lambda_i(a)a$ in B. For $p \in P = \prod_i t_i$ put $g_p = \prod_i p_i t^{e_i} \in \operatorname{Rees}(LB)$ and $f_p = \prod_i \lambda_i(p_i)p_i t^{e_i}$. Then $c = \sum_{p\in P} f_p$, and each f_p is homogeneous of degree d. It is relevant, since each $\lambda_i(p_i)p_i t^{e_i}$ divides f_p and the e_i generate $\mathbb{Z}^\iota$. The elements f_p/c of $\operatorname{Pot}\langle c\rangle$ sum to 1, so the opens $D(f_p/c) \cong \operatorname{Spec}\operatorname{Pot}\langle cf_p\rangle$ cover $\operatorname{Spec}\operatorname{Pot}\langle c\rangle$. Each maps to $\operatorname{Spec}\operatorname{Pot}\langle f_p\rangle$, and these maps glue to a morphism over $\operatorname{Spec} B$ from $\operatorname{Spec}\operatorname{Pot}\langle c\rangle$ to the scheme U glued from the $\operatorname{Spec}\operatorname{Pot}\langle f_p\rangle$, $p \in P$, along the Proj gluing data of $\operatorname{Rees}(LB)$ (`sum_open_finset`).

Step 4 (charts of $\mathrm{Bl}^{\mathrm{aff}}(L)$). Since g_p divides f_p, $\overline{\langle g_p\rangle} \subseteq \overline{\langle f_p\rangle}$, which gives a map $\operatorname{Pot}\langle g_p\rangle \to \operatorname{Pot}\langle f_p\rangle$. Let $\tilde{g}_p = \prod_i p_i t^{e_i} \in \operatorname{Rees}(L)$; the graded map $\operatorname{Rees}(L) \to \operatorname{Rees}(LB)$ sends $\tilde{g}_p$ to g_p and gives $\operatorname{Pot}\langle \tilde{g}_p\rangle \to \operatorname{Pot}\langle g_p\rangle$. The pairs (L_i, p_i), with $\Psi = s = \mathrm{id}$, form a presentation $F_p \in \mathcal{M}(L)$ (Definition 19), since $\mathcal{L}_i = L_i + (p_i) = L_i = L_{\Psi(i)}$ as $p_i \in L_i$. By the preliminary remark, $\operatorname{Pot}(S_{F_p}) = \operatorname{Pot}\langle \tilde{g}_p\rangle$. With the chart inclusion j_{F_p} of Definition 21 we obtain

$$h_p : \operatorname{Spec}\operatorname{Pot}\langle f_p\rangle \to \operatorname{Spec}\operatorname{Pot}\langle g_p\rangle \to \operatorname{Spec}\operatorname{Pot}(S_{F_p}) \xrightarrow{j_{F_p}} \mathrm{Bl}^{\mathrm{aff}}(L),$$

a morphism over $\operatorname{Spec} A$.

Step 5 (gluing on U). Let $\operatorname{Spec}\operatorname{Pot}(S)$ be the overlap of two charts of U. The image of c_i in $\operatorname{Pot}(S)$ is a non-zero-divisor (`ReesAlgebra.potion_algebraMap_mem_nonZeroDivisors`) and generates $L_i \operatorname{Pot}(S)$, so L becomes Cartier on $\operatorname{Spec}\operatorname{Pot}(S)$ (`loc_isCars_of_principal`). The restrictions of h_p and h_q to the overlap are both over $\operatorname{Spec} A$, so they agree by Proposition 23. Hence the h_p glue to $h : U \to \mathrm{Bl}^{\mathrm{aff}}(L)$ over $\operatorname{Spec} A$. Put $\varphi_\gamma : \operatorname{Spec} B \cong \operatorname{Spec}\operatorname{Pot}\langle c\rangle \to U \xrightarrow{h} \mathrm{Bl}^{\mathrm{aff}}(L)$; it is over $\operatorname{Spec} A$.

Step 6 (gluing on T). For γ, δ, the map $W = \operatorname{Spec} B_\gamma \times_T \operatorname{Spec} B_\delta \to T$ is an open immersion, hence flat (Section 4.1), so L becomes Cartier on W (`pullback_IsCars`). By Proposition 23, φ_γ and φ_δ agree on W. Hence the φ_γ glue to $\varphi : T \to \mathrm{Bl}^{\mathrm{aff}}(L)$ with $\varphi \circ u_\gamma = \varphi_\gamma$; it is over $\operatorname{Spec} A$. □

Putting Propositions 22, 23 and 24 together:
`Project/Blowups/BlowupExists.lean, lines 989--993, 1014--1018`

```
lemma ProjBlowup_UnivProp_existence_affine
    (A : CommRingCat) (L : ι→Ideal A) [fin : Fintype ι]
    {T : Scheme} [T.Over (Spec A)]
    (cond : pullback_Clos (T ↘Spec A) (loc_to_Clos A L) ∈CarsAsSubsetOfClos T) :
    ∃φ: T ⟶BlMu L, Scheme.Hom.IsOver φ(Spec A) := by

lemma ProjBlowup_UnivProp_affine
  (A : CommRingCat) (L : ι→Ideal A) [fin : Fintype ι]
  {T : Scheme} [T.Over (Spec A)]
  (cond : pullback_Clos (T ↘Spec A) (loc_to_Clos A L) ∈CarsAsSubsetOfClos T) :
    ∃! φ: T ⟶BlMu L, Scheme.Hom.IsOver φ(Spec A) := by
```

Theorem 25. $\mathrm{Bl}^{\mathrm{aff}}(L)$ *is terminal among* $\operatorname{Spec} A$*-schemes on which* L *becomes Cartier.*

The formalisation packages this as a structure, so that the affine blowup is an object carrying its own universal property rather than a scheme with three separate theorems attached.
`Project/Blowups/UniqueBlowup.lean, lines 38--47`

```
structure conceptual_blowup (Z : Clos X) where
  scheme : Scheme
  over : Scheme.Over scheme X
  in_cars : pullback_Clos (scheme ↘X) Z ∈CarsAsSubsetOfClos scheme
  φ(T : Scheme) [T.Over X] (in_preCars : pullback_Clos (T ↘X) Z ∈(CarsAsSubsetOfClos T)) :
    T ⟶scheme
  φ_over (T : Scheme) [T.Over X] (in_preCars : pullback_Clos (T ↘X) Z ∈(CarsAsSubsetOfClos
      T)) :
    Scheme.Hom.IsOver (φ T in_preCars) X
  φ_uniq (T : Scheme) [T.Over X] (in_preCars : pullback_Clos (T ↘X) Z ∈(CarsAsSubsetOfClos
      T)) :
    ∀φ' : T ⟶scheme, Scheme.Hom.IsOver φ' X →φ' = φT in_preCars
```

Theorem 25 is then the statement that $\mathrm{Bl}^{\mathrm{aff}}(L)$ carries this structure, for the centre L viewed as an element of $\mathrm{Clos}(\operatorname{Spec} A)$.
`Project/Blowups/BlowupExists.lean, lines 1042--1049`

```
def ProjBlowup_is_conceptual_blowups_affine (A : CommRingCat) (L : ι→Ideal A) [fin : Fintype ι]
    :
  conceptual_blowup (loc_to_Clos A L) where
  scheme := BlMu L
  over := inferInstance
  in_cars := blowups_Cars A L
  φT _ cond := ProjBlowup_φ A L cond
  φ_over T _ cond := ProjBlowup_φ_over A L cond
  φ_uniq T _ cond φ' hφ' := ProjBlowup_φ_uniq A L cond φ' hφ'
```

4.4 Base change along open immersions

To globalise we need to know how $\mathrm{Bl}^{\mathrm{aff}}$ behaves under restriction to an open.

Proposition 26. *Let $A \to B$ be a ring map with $\operatorname{Spec} B \to \operatorname{Spec} A$ an open immersion, and let $L' := (L_i B)_i$. Then there is exactly one isomorphism*

$$\mathrm{Bl}^{\mathrm{aff}}(L) \times_{\operatorname{Spec} A} \operatorname{Spec} B \;\cong\; \mathrm{Bl}^{\mathrm{aff}}(L')$$

over $\operatorname{Spec} B$.

`Project/Blowups/BlowupExists.lean, lines 1337--1341`

```
lemma base_change_Bl_open [Fintype ι] (A B : CommRingCat) [Algebra A B]
    [IsOpenImmersion (Spec B ↘Spec A)] (L : ι→Ideal A) :
    ∃! (e : pullback (BlMu L ↘Spec A) (Spec B ↘Spec A) ≅

    BlMu (L := fun i : ι=> Ideal.map (algebraMap A B) (L i))), Scheme.Hom.IsOver e.hom (Spec B)
        := by
```

Proof Put $P := \mathrm{Bl}^{\mathrm{aff}}(L) \times_{\operatorname{Spec} A} \operatorname{Spec} B$, with projections pr_1, pr_2. To show that L is Cartier on P, we apply a more general result, namely that the Cartier property is stable under flat base change (Section 4.1; `pullback_IsCars`): pr_1 is a base change of the open immersion

$\operatorname{Spec} B \to \operatorname{Spec} A$, hence flat, and L is Cartier on $\mathrm{Bl}^{\mathrm{aff}}(L)$ (Proposition 22). The pullback of L to $\operatorname{Spec} B$ is L' (`loc_to_Clos_baseChange`). Hence L' is Cartier on P, and, since L' is Cartier on $\mathrm{Bl}^{\mathrm{aff}}(L')$ (Proposition 22), L is Cartier on $\mathrm{Bl}^{\mathrm{aff}}(L')$. Theorem 25 gives $\varphi : P \to \mathrm{Bl}^{\mathrm{aff}}(L')$ over $\operatorname{Spec} B$ and $\psi : \mathrm{Bl}^{\mathrm{aff}}(L') \to \mathrm{Bl}^{\mathrm{aff}}(L)$ over $\operatorname{Spec} A$. The morphisms ψ and $\mathrm{Bl}^{\mathrm{aff}}(L') \to \operatorname{Spec} B$ have the same composite to $\operatorname{Spec} A$, hence define $\psi' : \mathrm{Bl}^{\mathrm{aff}}(L') \to P$ over $\operatorname{Spec} B$. Then $\mathrm{pr}_1 \circ \psi' \circ \varphi = \mathrm{pr}_1$ by the uniqueness in Theorem 25 for L on P over $\operatorname{Spec} A$, and $\mathrm{pr}_2 \circ \psi' \circ \varphi = \mathrm{pr}_2$ since φ is over $\operatorname{Spec} B$; hence $\psi' \circ \varphi = \mathrm{id}_P$. By the uniqueness for L' on $\mathrm{Bl}^{\mathrm{aff}}(L')$ over $\operatorname{Spec} B$, $\varphi \circ \psi' = \mathrm{id}$. By the uniqueness for L' on P over $\operatorname{Spec} B$, any isomorphism $P \to \mathrm{Bl}^{\mathrm{aff}}(L')$ over $\operatorname{Spec} B$ equals φ. □

4.5 The local blowups and their overlaps

Now let Z be a presented centre on X, with cover $\mathcal{U} = (u_\gamma : U_\gamma = \operatorname{Spec} A_\gamma \to X)_{\gamma \in J}$ and ideals $I_{i\gamma}$. For each γ set

$$B_\gamma \;:=\; \mathrm{Bl}^{\mathrm{aff}}\big((I_{i\gamma})_i\big),$$

the affine blowup of the chart, a scheme over $\operatorname{Spec} A_\gamma$ and hence over X.

For a pair γ, δ let $U_{\gamma\delta} := U_\gamma \times_X U_\delta$ be the overlap, an open of both charts, and let

$$V_{\gamma\delta} \;:=\; B_\gamma \times_{\operatorname{Spec} A_\gamma} U_{\gamma\delta}$$

be its preimage in B_γ. This is an open subscheme of B_γ, and we write $f_{\gamma\delta} : V_{\gamma\delta} \hookrightarrow B_\gamma$ for the inclusion and $\pi_{\gamma\delta} : V_{\gamma\delta} \to U_{\gamma\delta}$ for the projection.
`Project/Blowups/BlowupExists.lean, lines 1393--1400, 1417--1418, 1437--1441, 1614--1620`

```
def ideal_loc (X : Scheme) (Z : PreClos X) (γ : Z.cov.J) : Z.indnumb →Ideal (Z.cov.obj γ) :=
  fun (i : Z.indnumb) => Z.ideal i γ

def Proj_loc (X : Scheme) (Z : PreClos X) (γ : Z.cov.J)
    [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] : Scheme :=
  BlMu (A := Z.cov.obj γ) (ideal_loc X Z γ)

def open_pair (X : Scheme) (Z : PreClos X) (γ δ: Z.cov.J) : Scheme :=
  pullback (Z.cov.map γ) (Z.cov.map δ)

def Proj_loc_pair (X : Scheme) (Z : PreClos X) (γ δ: Z.cov.J)
  [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] : Scheme :=
    pullback (pullback.fst (Z.cov.map γ) (Z.cov.map δ))

def Proj_loc_pair_mor (X : Scheme) (Z : PreClos X) (γ δ: Z.cov.J)
    [DecidableEq Z.indnumb]
    [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
  Proj_loc_pair X Z γδ⟶open_pair X Z γδ:=
    pullback.fst (pullback.fst (Z.cov.map γ) (Z.cov.map δ))
      (Proj_loc X Z γ↘Spec (Z.cov.obj γ))
```

Here `Proj_loc` is B_γ, `open_pair` is $U_{\gamma\delta}$, `Proj_loc_pair` is $V_{\gamma\delta}$, and `Proj_loc_pair_mor` is $\pi_{\gamma\delta}$. The inclusion $f_{\gamma\delta}$ is the other projection of the same fibre product.

Proposition 27. *There is exactly one isomorphism*

$$t_{\gamma\delta} : V_{\gamma\delta} \xrightarrow{\sim} V_{\delta\gamma}$$

over $U_{\gamma\delta}$.

This is the heart of the globalisation, and it is again an application of affine terminality, now on the overlap. Both $V_{\gamma\delta}$ and $V_{\delta\gamma}$ sit over $U_{\gamma\delta}$. Locally on $U_{\gamma\delta}$, say on an affine open $\operatorname{Spec} C \subseteq U_{\gamma\delta}$, Proposition 26 identifies

$$V_{\gamma\delta} \times_{U_{\gamma\delta}} \operatorname{Spec} C \;\cong\; \mathrm{Bl}^{\mathrm{aff}}\big(I_{i\gamma}C\big), \qquad V_{\delta\gamma} \times_{U_{\gamma\delta}} \operatorname{Spec} C \;\cong\; \mathrm{Bl}^{\mathrm{aff}}\big(I_{i\delta}C\big).$$

The two centres appearing on the right are equal, because both are the restriction to $\operatorname{Spec} C$ of the global centre Z: the composites $\operatorname{Spec} C \to U_\gamma \to X$ and $\operatorname{Spec} C \to U_\delta \to X$ agree, being the two projections out of a fibre product over X. So the two sides are the affine blowup of one and the same centre, and Theorem 25 gives a unique isomorphism between them over $\operatorname{Spec} C$.

Uniqueness at each $\operatorname{Spec} C$ is what allows these local isomorphisms to be glued into $t_{\gamma\delta}$, and it also makes $t_{\gamma\delta}$ itself unique. The statement proved in Lean records both the map and its restriction behaviour.

`Project/Blowups/BlowupExists.lean, lines 1963--1973`

```
def Proj_loc_pair_lemm (X : Scheme) (Z : PreClos X) (γ δ: Z.cov.J)
  [DecidableEq Z.indnumb]
  [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
  ∃! (f : (Proj_loc_pair X Z γδ) ⟶(Proj_loc_pair X Z δγ)),

  (∃ (pf : Scheme.Hom.IsOver f (open_pair X Z γδ)),
    ∀(C : CommRingCat) (i : Spec C ⟶open_pair X Z γδ) [IsOpenImmersion i],
      restrictToOpen f i =
      (pullbackSymmetry _ _).hom ≫Proj_loc_pair_open_φ X Z γδC i ≫
      (pullbackSymmetry _ _).hom) := by
```

The morphism $t_{\gamma\delta}$ itself is then extracted from that unique existence statement.

`Project/Blowups/BlowupExists.lean, lines 2259--2264`

```
def Proj_loc_pair_swap (X : Scheme) (Z : PreClos X) (γ δ: Z.cov.J)
  [DecidableEq Z.indnumb]
  [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    (Proj_loc_pair X Z γδ) ⟶(Proj_loc_pair X Z δγ) :=
  Classical.choose (Proj_loc_pair_lemm X Z γδ)
```

4.6 Triple overlaps and the cocycle

To glue the B_γ along the $t_{\gamma\delta}$ we must supply the triple overlap data and verify the cocycle condition. Let

$$P_{ijk} \;:=\; V_{ij} \times_{B_i} V_{ik},$$

the preimage in B_i of $U_i \cap U_j \cap U_k$. The required map is

$$t'_{ijk} : P_{ijk} \longrightarrow P_{jki},$$

compatible with t_{ij}, and the cocycle condition is

$$t'_{ijk} \circ t'_{jki} \circ t'_{kij} = \mathrm{id}.$$

The usual construction of t'_{ijk} is topological: one checks that the image of the triple overlap under t_{ij} lies inside the double overlap V_{jk}, and restricts. Here this is not needed, and the reason is worth spelling out because it is the cleanest illustration of how the universal properties organise the proof.

Recall that V_{jk} was defined as a fibre product, $V_{jk} = B_j \times_{\operatorname{Spec} A_j} U_{jk}$, and that U_{jk} is itself a fibre product $U_j \times_X U_k$. So to give a map $P_{ijk} \to V_{jk}$ it is enough to give three maps

$$P_{ijk} \to B_j, \qquad P_{ijk} \to \operatorname{Spec} A_j, \qquad P_{ijk} \to \operatorname{Spec} A_k,$$

agreeing over $\operatorname{Spec} A_j$ and over X respectively. All three are available. The first is

$$P_{ijk} \xrightarrow{\mathrm{pr}_1} V_{ij} \xrightarrow{t_{ij}} V_{ji} \hookrightarrow B_j,$$

the second is its composite with $B_j \to \operatorname{Spec} A_j$, and the third is

$$P_{ijk} \xrightarrow{\mathrm{pr}_2} V_{ik} \xrightarrow{t_{ik}} V_{ki} \hookrightarrow B_k \to \operatorname{Spec} A_k.$$

The compatibility over X holds because t_{ij} and t_{ik} are morphisms over the respective overlaps, so both composites $P_{ijk} \to X$ agree with the structure morphism. So t'_{ijk} is built by the universal property of fibre products alone, with no reference to underlying topological spaces.

`Project/Blowups/BlowupExists.lean, lines 2433--2446`

```
noncomputable def Proj_loc_pair_t' (X : Scheme) (Z : PreClos X) (i j k : Z.cov.J)
    [DecidableEq Z.indnumb] [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    pullback (Proj_loc_pair_incl X Z i j) (Proj_loc_pair_incl X Z i k) ⟶
      pullback (Proj_loc_pair_incl X Z j k) (Proj_loc_pair_incl X Z j i) :=
  pullback.lift
    (pullback.lift
      (pullback.lift
        (pullback.fst (Proj_loc_pair_incl X Z i j) (Proj_loc_pair_incl X Z i k) ≫
          Proj_loc_pair_swap X Z i j ≫Proj_loc_pair_incl X Z j i ≫
          (Proj_loc X Z j ↘Spec (Z.cov.obj j)))
        (pullback.snd (Proj_loc_pair_incl X Z i j) (Proj_loc_pair_incl X Z i k) ≫
          Proj_loc_pair_swap X Z i k ≫Proj_loc_pair_incl X Z k i ≫
          (Proj_loc X Z k ↘Spec (Z.cov.obj k)))
```

The three nested `pullback.lift`s are exactly the three maps above, assembled into $V_{jk} = B_j \times_{\operatorname{Spec} A_j} (U_j \times_X U_k)$ from the inside out.

For the cocycle, note that all the inclusions $f_{ij} : V_{ij} \to B_i$ are open immersions, hence monomorphisms, so $P_{ijk} \to B_i$ is a monomorphism too and it suffices to prove

$$\left(t'_{ijk} \circ t'_{jki} \circ t'_{kij}\right) \circ m_{ijk} \;=\; m_{ijk}, \qquad m_{ijk} : P_{ijk} \hookrightarrow B_i.$$

Both sides are morphisms $P_{ijk} \to B_i = \mathrm{Bl}^{\mathrm{aff}}\big((I_{\bullet i})\big)$ from an open subscheme of B_i. By Section 4.1 the centre is Cartier on that open, so Proposition 23 applies: it is enough to check the two morphisms agree after composing with $B_i \to \operatorname{Spec} A_i$. Tracing through the definitions, that computation telescopes and reduces to the single identity

$$t_{ij} \circ t_{ji} = \mathrm{id}_{V_{ij}},$$

which is proved the same way, by the same uniqueness statement, using that a double swap induces the identity on U_{ij} since it is the composite of the two symmetry isomorphisms of a fibre product.

So the entire triple overlap analysis reduces to one instance of affine uniqueness. No intersections of opens are ever computed.

4.7 Gluing and the global universal property

We can now assemble the gluing datum. Recall what has to be supplied: objects B_γ, opens $V_{\gamma\delta} \hookrightarrow B_\gamma$ with $V_{\gamma\gamma} = B_\gamma$, transition isomorphisms $t_{\gamma\delta} : V_{\gamma\delta} \to V_{\delta\gamma}$ with $t_{\gamma\gamma} = \mathrm{id}$, triple overlap maps $t'_{\gamma\delta\epsilon}$ compatible with the $t_{\gamma\delta}$, and the cocycle identity. We have produced all of these:

- B_γ and $V_{\gamma\delta}$ in Section 4.5;
- $t_{\gamma\delta}$ in Proposition 27;
- $t'_{\gamma\delta\epsilon}$ and the cocycle in Section 4.6.

The two degenerate conditions are worth a comment. That $f_{\gamma\gamma}$ is an isomorphism holds because $U_{\gamma\gamma} = U_\gamma \times_X U_\gamma$ equals U_γ, since u_γ is a monomorphism. That $t_{\gamma\gamma} = \mathrm{id}$ then follows from the uniqueness clause of Proposition 27, since the identity is a morphism over $U_{\gamma\gamma}$ and there is only one such. Both are places where an informal account would not pause, and both need the monomorphism property of open immersions to make the two available structure morphisms on $V_{\gamma\gamma}$ agree.

The data $\big(B_\gamma, V_{\gamma\delta}, f_{\gamma\delta}, t_{\gamma\delta}, t'_{\gamma\delta\epsilon}\big)$ is therefore a gluing datum in the sense of [12], and we define $\mathrm{Bl}_Z X$ to be the resulting scheme, with its morphism to X induced by the $B_\gamma \to \operatorname{Spec} A_\gamma \to X$. The canonical maps $\iota_\gamma : B_\gamma \to \mathrm{Bl}_Z X$ are open immersions whose images cover, and they satisfy

$$\iota_\gamma \circ f_{\gamma\delta} \;=\; \iota_\delta \circ f_{\delta\gamma} \circ t_{\gamma\delta}$$

on $V_{\gamma\delta}$, which is the identity used in the gluing arguments below. The glued scheme is the following, `PreBlGlob` being the gluing datum whose fields are the constructions above.

`Project/Blowups/BlowupExists.lean, lines 2586--2589`

```
abbrev BlGlob (Z : PreClos X) [DecidableEq Z.indnumb]
  [Fintype Z.indnumb]
  [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] : Scheme :=
  Scheme.GlueData.glued (PreBlGlob Z)
```

Proposition 28. *The centre pulled back to* $\mathrm{Bl}_Z X$ *is Cartier.*

The presentation is assembled from the affine ones. Its cover is the family of all dilatation charts of all the local blowups,

$$\big\{\operatorname{Spec} A_\gamma[F] \;:\; \gamma \in J,\; F \in \mathcal{M}(I_{\bullet\gamma})\big\},$$

indexed by the pairs (γ, F), each mapping into $\operatorname{Bl}_Z X$ as the composite

$$\operatorname{Spec} A_\gamma[F] \hookrightarrow B_\gamma \xrightarrow{\iota_\gamma} \operatorname{Bl}_Z X,$$

a composite of open immersions and hence an open immersion. These charts cover, since the ι_γ cover $\operatorname{Bl}_Z X$ and the dilatation charts cover each B_γ.

The ideals are the ones supplied by Proposition 22, namely $I_{i\gamma} \cdot A_\gamma[F]$, and they are principal on non-zero-divisors for the same reason as there. What has to be checked in addition is that these ideals really do present the pulled back centre, that is, that

$$\operatorname{Spec}\big(A_\gamma[F]/I_{i\gamma}A_\gamma[F]\big) \;\cong\; Z_i \times_X \operatorname{Spec} A_\gamma[F],$$

compatibly with the structure morphisms. This follows by composing the corresponding isomorphism for the presented centre Z over U_γ with the base change of $Z_i \times_X U_\gamma$ along $\operatorname{Spec} A_\gamma[F] \to \operatorname{Spec} A_\gamma$, using that a fibre product taken in two stages agrees with the one taken in a single stage. In the formalisation this composite is written out explicitly, and it is where most of the length of the construction sits.

The Cartier conditions themselves are the affine ones, unchanged. This gives Theorem 7.

`Project/Blowups/BlowupExists.lean, lines 3294--3300`

```
def BlGlobPreClos (Z : PreClos X)
    [DecidableEq Z.indnumb] [Fintype Z.indnumb]
    [(i : Z.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z.indnumb))] :
    PreClos (BlGlob Z) where
  indnumb := Z.indnumb
  subscheme i := pullback (Z.subscheme i ↘X) ((BlGlob Z) ↘X)
  over i := ⟨pullback.snd _ _⟩
```

The choice of cover here is the whole content. A cover of $\operatorname{Bl}_Z X$ by arbitrary affine opens would not do, since the pulled back centre is invertible but not necessarily principal on such an open. It is because the Cartier condition of Definition 4 quantifies existentially over presentations that we are free to use the cover we built the scheme from, on which principality holds by construction.

For the universal property, let $T \to X$ be such that the centre becomes Cartier on T. Set $T_\gamma := T \times_X U_\gamma$. On T_γ the centre $I_{\bullet\gamma}$ becomes Cartier, by restricting the given presentation, so Theorem 25 gives a unique

$$\varphi_\gamma : T_\gamma \longrightarrow B_\gamma$$

over $\operatorname{Spec} A_\gamma$. Composing with $B_\gamma \to \operatorname{Bl}_Z X$ gives maps $T_\gamma \to \operatorname{Bl}_Z X$ over X, and these glue: on $T_\gamma \times_T T_\delta$ both restrictions are morphisms over the overlap into a blowup of a Cartier centre, so they agree by Proposition 23. Note again that the compatibility

is not computed but deduced from uniqueness. This produces $\varphi : T \to \mathrm{Bl}_Z X$ over X, and gives Theorem 5's existence half.

Uniqueness is the same argument with the maps given rather than constructed. If φ, φ' are two morphisms $T \to \mathrm{Bl}_Z X$ over X, they agree after restriction to each T_γ by Proposition 23, hence agree.

4.8 Independence of the presentation

It remains to see that $\mathrm{Bl}_Z X$ depends only on the centre and not on the presentation, which is what allows Theorem 6 to be stated for $\mathcal{Z} \in \mathrm{Clos}(X)$.

Proposition 29. *If Z' and Z'' are equivalent presented centres, then there is exactly one isomorphism $\mathrm{Bl}_{Z'} X \cong \mathrm{Bl}_{Z''} X$ over X.*

`Project/Blowups/BlowupExists.lean, lines 3354--3361`

```
lemma PreProjBlowup_rel (Z' Z'' : PreClos X) (eq : Quotient.mk' Z' = Quotient.mk' Z'')
    [DecidableEq Z'.indnumb]
    [Fintype Z'.indnumb]
    [Fintype Z''.indnumb]
    [DecidableEq Z''.indnumb]
    [(i : Z'.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z'.indnumb))]
    [(i : Z''.indnumb →₀ ℤ) →Decidable (i ∈Set.range (ρNatToInt Z''.indnumb))] :
  ∃! (e : BlGlob Z' ≅BlGlob Z''), Scheme.Hom.IsOver e.hom X := by
```

Once Theorem 7 is available this is formal, and it is the same two-sided argument as Proposition 26. Since Z' and Z'' represent the same element of $\mathrm{Clos}(X)$, the centre that becomes Cartier on $\mathrm{Bl}_{Z'} X$ is the same element of Clos as the one that becomes Cartier on $\mathrm{Bl}_{Z''} X$. So Theorem 5 applied to Z'' with $T = \mathrm{Bl}_{Z'} X$ gives a unique map one way, the symmetric application gives a unique map the other way, and both composites are identities by uniqueness.

This uses the quotient $\mathrm{Clos}(X)$. The universal property is applied in the form where the Cartier hypothesis is stated for the class $[Z] \in \mathrm{Clos}(X)$ (`PreProjBlowup_UnivProp_existence_Cars`, `PreProjBlowup_UnivProp_unicity_Cars`). Since $[Z'] = [Z'']$, the hypotheses for Z' and for Z'' coincide, so no comparison of the covers of Z' and Z'' is needed.

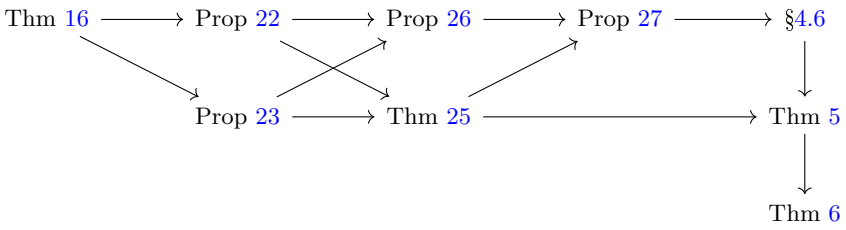
We may therefore define, for a centre $\mathcal{Z} \in \mathrm{Clos}(X)$,

$$\mathrm{Bl}_{\mathcal{Z}} X \;:=\; \mathrm{Bl}_Z X \quad \text{for any presentation } Z \text{ of } \mathcal{Z},$$

well defined up to unique isomorphism over X, and Theorem 6 follows from Theorem 5 by rewriting the Cartier hypothesis along the equality $[\,Z\,] = \mathcal{Z}$. This completes the proof.

4.9 Dependencies

The logical structure of the proof is the following, read left to right.



Everything originates in the universal property of the dilatation. The affine terminality statement, Theorem 25, is used four separate times downstream: to build the base change isomorphism, to build the overlap comparison, to close the cocycle, and to glue the local maps into the global one. Proposition 23, the uniqueness half, is used more often still, since it is what makes each of those constructions canonical and hence compatible with the others.

Note that the diagram has no cycles and no step that depends on the global object being already constructed. This is not automatic. A common way to organise a gluing argument is to construct the object first and verify its properties afterwards, which works informally because one can speak of the object before knowing that it is well defined. Here every arrow points forward, and the global scheme $\mathrm{Bl}_Z X$ is named only after the gluing datum has been completely verified.

5 Correspondence

For reference, the statements of this paper and the declarations proving them.

Statement	Lean declaration	File
Def. 2	`PreClos`	`PreClosAndClos.lean`
Def. 3	`rel`, `Clos`	`PreClosAndClos.lean`
Def. 4	`IsPreCars`, `IsCars`	`PreClosAndClos.lean`
§2.2	`pullback_PreClos`, `pullback_Clos`	`PreClosAndClos.lean`
Def. 12	`Multicenter`, `LargeIdeal`	`Multicenter.lean`
Def. 13	`PreDil`, `r`, `Dilatation`	`Multicenter.lean`
Lem. 14	`nonzerodiv_image`	`Multicenter.lean`
Lem. 15	`image_elem_LargeIdeal_equal`	`Multicenter.lean`
Thm. 16	`desc`, `lemma_exists_unique_morphism`	`Multicenter.lean`
Cor. 17	`ofFamilyIso`	`Multicenter.lean`
Prop. 18	`functo_dila_alg`	`Multicenter.lean`
Def. 19	`Mu`	`Bl.lean`
Prop. 20	`clo_mu_mor`, `Mu_mor_iso`	`Bl.lean`
Def. 21	`BlMu`	`Bl.lean`
Prop. 22	`BlMuPreClos_IsPreCars`, `blowups_Cars`	`Cars.lean`
Prop. 23	`ProjBlowup_UnivProp_unicity_affine`	`UniqueBlowup.lean`
Prop. 24	`ProjBlowup_UnivProp_existence_affine`	`BlowupExists.lean`
Thm. 25	`ProjBlowup_is_conceptual_blowups_affine`	`BlowupExists.lean`
Prop. 26	`base_change_Bl_open`	`BlowupExists.lean`
§4.5	`Proj_loc`, `Proj_loc_pair`	`BlowupExists.lean`
Prop. 27	`Proj_loc_pair_lemm`	`BlowupExists.lean`
§4.6	`Proj_loc_pair_t'`, `PreBlGlob`	`BlowupExists.lean`
§4.7	`BlGlob`	`BlowupExists.lean`
Prop. 28	`BlGlobPreClos`, `BlGlob_IsCars`	`BlowupExists.lean`
Thm. 5	`PreProjBlowup_UnivProp`	`BlowupExists.lean`
Prop. 29	`PreProjBlowup_rel`	`BlowupExists.lean`
Thm. 6	`GlobalBlowup_UnivProp`	`BlowupExists.lean`

Files in `Project/Blowups/` except `Multicenter.lean`, which is in `Project/Dilatation/`.

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